Assessment Tool-2
CSA1507 cloud computing
with big data analysis

1) Cloud computing:

A * Delivery of computing service over the internet

* characteristics

* on demand self service
* Broad network acess
* Resource pooling
* Rapid elasticity
* measured service

Benifits

* cost- effective
* scalability
* High availability
* Flexibility
* Automatic update
* Disable recovery

limitations

* Security concerns
* privacy issues
* Internet dependency
* Downtime
* vendor lock-in

Concept map: cloud computing

Cloud computing

Definition
* Internet
* Service
* Resource

Characteristics
* on demand
* scalability
* Resource Pool
* Elasticity

Benefits
* Low cost
* Flexible
* Reliable
* Available

Limitations
* Security
* Privacy
* Downtime
* Vendor lock-in

2A)

Concept map 2: Evolution of cloud computing

-> **Hardware Evolution**

* Mainframe computers
* personal computers
* servers
* virtualization

-> **Internet & software Evolution.**

* Internet
* web Applications
* service oriented Architecture (SOA)
* API's

-> **Distributed systems**

* Grid computing
* cluster computing
* utility computing

Service Delivery

* IaaS
* PaaS
* SaaS
* XaaS

Result

Modern cloud computing.

concept map 2: ... computing

Evolution of cloud

- Hardware Evolution
  - * main frame
  - * PC
  - * servers
  - * virtualize
- Internet & software
  - * Internet
  - * web APPS
  - * SOA
  - * API's
- Distributed systems
  - * Grid
  - * cluster
  - * utility
  - * comp.
- service delivery
  - * Iaas
  - * paas
  - * saas
  - * Xaas

Modern cloud computing

concept map3: service models.

3) concept
cloud service models

Iaas (Infrastructure as a service)

provider Responsibilites

* servers
* storage
* networking
* virtualization

user Responsibilities

* operating system
* Applications
* Data
* security settings

paas (platform ...

provider Responsibilities

* Infrastructure
* platform
* Application
* maintenance

user Responsibilities

* use the software
* Manage user data.

↓

Xaas (Anything as a service)

Examples:

* DBaas
* STaas
* Naas
* security as a service.

provider delivers specialized service over the internet.

4) Concept Map 4: virtualization and web services as Enablers of cloud

cloud

↓

cloud computing

virt

* Hypervisor
* virtual Machines
* Resource sharing
* server consolidation
* scalability
* cost Reduction.

↓

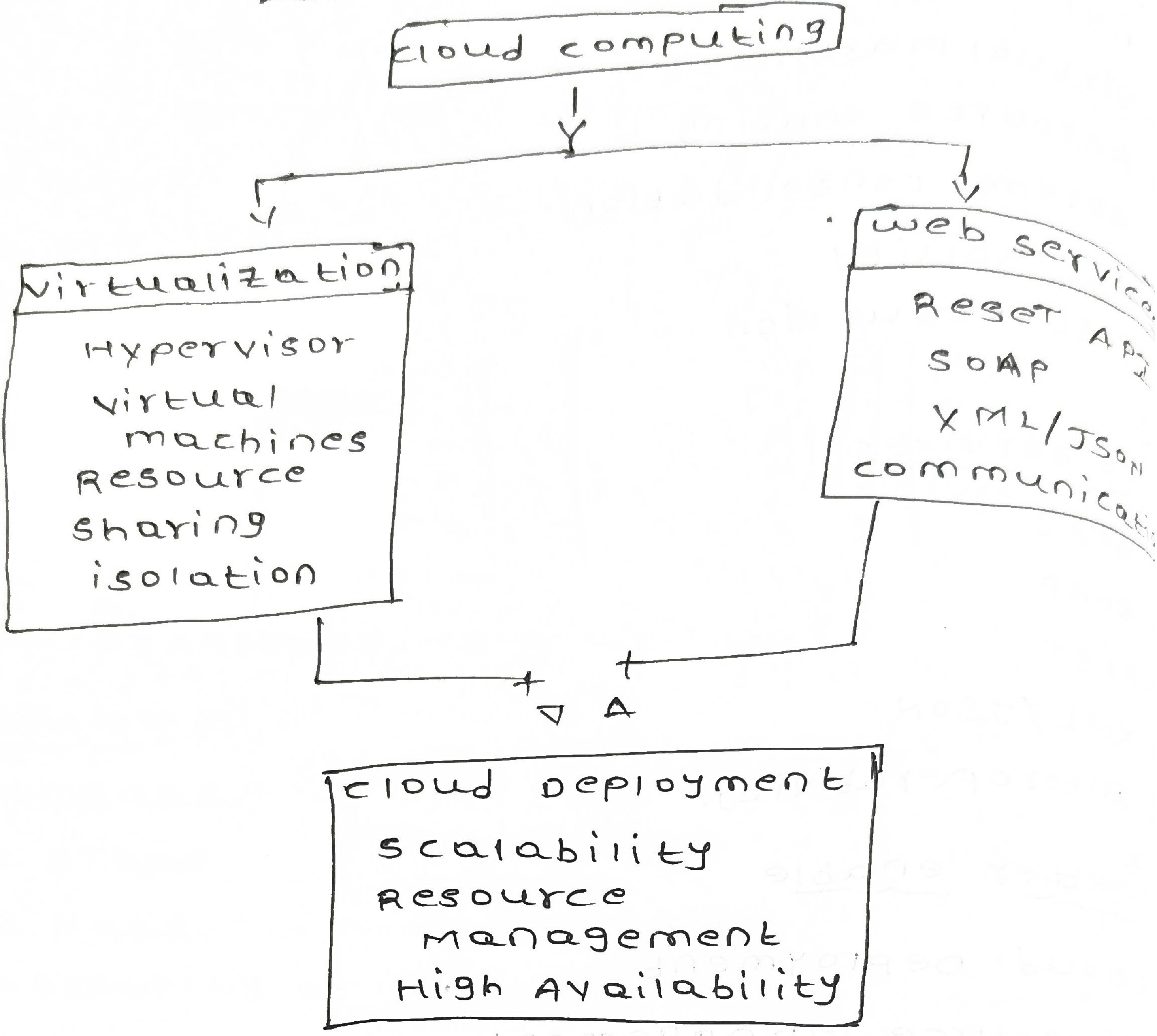
web services

* API's
* SOAP
* REST
* XML / JSON
* Interoperability.

↓

Together enable

* cloud Deployment
* Resource Management
* scalability
* flexibility
* High Availability
* Remote acess


cloud computing
virtualization
Hypervisor
virtual machines
Resource sharing
isolation
web service
Reset API
SOAP
XML/JSON
communication
cloud Deployment
scalability
Resource management
High Availability